Diego Fernando Mancera Gómez

Full-Stack Web Developer | Linux Kernel Contributor

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SUMMARY

Full-Stack Web Developer and Computer Systems Engineering student with 3 years of freelance experience building web applications with Next.js and Supabase (PostgreSQL). Linux kernel contributor with two security vulnerability fixes (CVEs) merged upstream, and open source contributor to HAMi, a Kubernetes GPU scheduler (CNCF).

SKILLS

Programming Languages: JavaScript, TypeScript, Python, C, Go, Dart, SQL
Frontend: Next.js, HTML, CSS, Flutter
Backend: REST APIs, Supabase, Firebase Auth, Gemini API
Databases: PostgreSQL, Firebase
DevOps and Tools: Git, GitHub, GitHub Actions, CI/CD, Vercel, Linux, Kubernetes
Systems: Linux Kernel, Device Drivers, Debugging, KASAN, syzbot
AI: Claude Code, Codex, Cursor, MCP, Prompt Engineering, Local LLMs (Ollama)

EXPERIENCE

Full-Stack Web Developer | CapiHw Labs (Freelance), León, Mexico | 2023 – Present

- Delivered **10+ production websites** for clients in marketing, design, technology and construction, **5 of them with a Supabase (PostgreSQL) backend**.
- Built a **product catalog website for Betterware products** with Next.js and Supabase.
- Built a **sales management system** with Supabase that tracks orders, payments and credit sales per customer.
- Automated deployments with GitHub Actions and Vercel, **reducing deployment time from 10 to 4–5 minutes**.
- Debugged, tested and fixed every release myself, working on Git feature branches to keep main stable; used AI coding agents (Claude Code, Codex) to speed up development.

OPEN SOURCE CONTRIBUTIONS

Linux Kernel Contributor (C) | Remote | Aug 2025 – Present

- Fixed an out-of-bounds memory read (CVE-2026-68344) in the ueagle-atm USB driver, detected with KASAN and syzbot, that a malicious USB device could trigger. Merged in Linux 7.2 and backported to stable kernels 5.10 to 7.1.
- Fixed a use-after-free bug (CVE-2026-89892) in the em28xx media driver reported by syzbot. Merged in Linux 7.3 and backported to stable kernels 5.10 to 7.2.

HAMi Contributor, Kubernetes GPU Scheduler, CNCF (Go) | Remote | Sep 2026 – Present

- Fixed a scheduler scoring bug (PR #3106) that calculated pod memory 59% lower than the reserved amount and ranked GPUs incorrectly; added Go regression tests.
- Fixed a bug (PR #3313) in the Enflame backend that read a mode value as a memory percentage.

PROJECTS

Cuentivo, Android App for Credit Sales | Flutter, Dart, Firebase (Blaze plan), Gemini API

- Mobile app for small businesses to track sales, payments and overdue balances per customer. Tested by 20 users with 10+ confirmed installs.
- Pro version uses Firebase Auth and the Gemini API through a secure backend gateway.
- Ran AI-assisted security testing to find vulnerabilities and fixed them in the architecture.

Local LLM Setup | Ollama, Linux, Windows

- Installed and ran local language models (Gemma, DeepSeek-Coder-V2, Qwen) on Linux and Windows, choosing lightweight models that fit an RTX 5060 GPU with 8 GB of VRAM.

PDF Crop Studio, PDF Image Extractor | Python, Open Source
Crops images from PDF files with pixel accuracy, auto-labels them from the text layer and falls back to OCR.
github.com/diegodevxd/pdf-crop-studio

EDUCATION

B.S. in Computer Systems Engineering | Universidad EPCA, León, Mexico | 2024 – Present (graduating 2028)

CERTIFICATIONS

- LFD103: A Beginner's Guide to Linux Kernel Development, The Linux Foundation (2026)
- LFC102: Inclusive Open Source Community Orientation, The Linux Foundation (2026)

LANGUAGES

Spanish (Native), English (B2, Upper Intermediate)